Terms, Factors, Coefficients, and Degree
Answer Key
Algebra 1

Quick Answers

1. $x^2 + 5x$
2. $x^2 + 8x + 15$
3. $4x^3 - x^2 + 6x - 9$
4. $(x + 3)(x + 4)$
5. $6x + 15$
6. $7x + 3$
7. $2x^2 + 5x - 12$
8. $3x^2(2x - 5)$
9. $2x^5 + 7x^3 - 4x$
10. $x^2 + 8x + 12$

Curriculum

Level: Algebra 1
HSA-APR.A.1 – *problems 1, 2, 3, 5, 6, 7, 9, 10*
HSA-SSE.B.3 – *problems 4, 8*
Difficulty 1–5 of 5 across 11 tagged checks.

1. Area model for width x, length $x + 5$.
Solution: The two cells are $x \cdot x = x^2$ and $x \cdot 5 = 5x$, so the area is the sum of those two cells: $\boxed{x^2 + 5x}$
Reading the answer: two terms. The coefficient of x is 5; the coefficient of x^2 is 1 (an invisible 1 is still a coefficient). Degree 2, because the largest exponent among the terms is 2.

2. Model for $(x + 3)(x + 5)$.
Solution: cells x^2, $5x$, $3x$, 15. The two middle cells are like terms ($5x + 3x = 8x$), so four cells collapse to three terms: $\boxed{x^2 + 8x + 15}$
Why fewer terms: $5x$ and $3x$ have the same variable to the same power, so they combine. Cells count multiplications; terms count what is left after combining like terms.

3. Terms out of order: -9, $6x$, $4x^3$, $-x^2$.
Solution: coefficient and degree of each term — -9: coefficient -9, degree 0 (a constant is $-9x^0$); $6x$: coefficient 6, degree 1; $4x^3$: coefficient 4, degree 3; $-x^2$: coefficient -1, degree 2. Standard form orders the terms by decreasing degree: $\boxed{4x^3 - x^2 + 6x - 9}$
The polynomial has degree 3 (the largest term degree) and leading coefficient 4 (the coefficient of the highest-degree term, not the number written first in the original list).

4. Factor $x^2 + 7x + 12$.
Solution: find two numbers with product 12 and sum 7: 3 and 4. $\boxed{(x + 3)(x + 4)}$
Degrees: each factor has degree 1, and the product has degree $1 + 1 = 2$ — multiplying polynomials adds their degrees.

5. $3(2x+5)$ — sum of terms, or product of factors?
Solution: Ann is describing it as written: $3(2x+5)$ is a single term built as a product of the factors 3 and $(2x+5)$. Distributing rewrites it as a sum:

$$3(2x+5) = 3\cdot 2x + 3\cdot 5 = 6x+15$$

$\boxed{6x+15}$
Factored form: factors 3 and $(2x+5)$; one term. Expanded form: terms $6x$ and 15; no common factor written out. Jo is right about the expanded form only.

6. Add $(3x^2+5x-1)+(-3x^2+2x+4)$.
Solution: combine like terms by degree: $3x^2+(-3x^2)=0, \quad 5x+2x=7x, \quad -1+4=3.$
$\boxed{7x+3}$
What happened to the degree: the two degree-2 terms are opposites, so the x^2 term has coefficient 0 and disappears. The sum has degree 1. Adding polynomials can lower the degree; it can never raise it.

7. Model for $(2x-3)(x+4)$.
Solution: cells $2x\cdot x = 2x^2, \quad 2x\cdot 4 = 8x, \quad -3\cdot x = -3x, \quad -3\cdot 4 = -12$. Combine $8x+(-3x)=5x$: $\boxed{2x^2+5x-12}$
The circled cell is the bottom-right one, $-3\cdot 4=-12$: the constant term always comes from multiplying the two constant parts. The coefficient of x is 5.

8. Factor $6x^3-15x^2$ completely.
Solution: the greatest common factor of $6x^3$ and $15x^2$ is $3x^2$ (3 divides 6 and 15; x^2 is the highest power in both):

$$6x^3-15x^2 = 3x^2(2x-5)$$

$\boxed{3x^2(2x-5)}$
Factors: 3 (degree 0), x^2 (degree 2), and $(2x-5)$ (degree 1). Degrees add: $0+2+1=3$, matching the degree of the original polynomial.

9. Multiply $(2x^3-x)(x^2+4)$.
Solution:

$$\begin{aligned}(2x^3-x)(x^2+4) &= 2x^3\cdot x^2 + 2x^3\cdot 4 - x\cdot x^2 - x\cdot 4\\ &= 2x^5+8x^3-x^3-4x\end{aligned}$$

Combine the like x^3 terms: $8x^3-x^3=7x^3$. $\boxed{2x^5+7x^3-4x}$
Degree and leading coefficient: degree 5, leading coefficient 2. Predictable without multiplying everything: the highest-degree term of the product is the product of the two highest-degree terms, $2x^3\cdot x^2=2x^5$, so the degree is $3+2=5$ and the leading coefficient is $2\cdot 1=2$.

10. Is $x^2 + 8x + 12$ equivalent to $(x + 2)(x + 6)$?
Solution: expand the product — two numbers with product 12 and sum 8 are 2 and 6:

$$(x + 2)(x + 6) = x^2 + 6x + 2x + 12 = x^2 + 8x + 12$$

$\boxed{x^2 + 8x + 12}$ so the two expressions ARE equivalent.
Explanation (accept any equivalent reasoning; read the student's words). "Sum of three terms" and "product of two factors" describe two *forms* of writing, not two different expressions. The same number comes out for every value of x, so it is one expression with two useful forms: the factored form shows the zeros ($x = -2$ and $x = -6$), the expanded form shows the degree, the leading coefficient, and the constant term. Maya is confusing how an expression is written with what it is.